THE TRAINING DATA LESSON · LESSON 2 OF 6

# What’s in the Playlist?

How AI learns — and where bias comes from. An AI is a music app, its training data is the playlist, and a one-vibe playlist (all loud, all fast, all party) means one-vibe answers. Stands alone.

## AT A GLANCE

| TIME | MATERIALS | TECH NEEDED | PREP |
|---|---|---|---|
| 45 min | Playlist Check worksheet, 2 pages, 1 per student — each fills their OWN sheet (typeable PDF copy included) · Slides (optional; script in the notes) · board · scissors optional | None — runs fully unplugged. Optional: a projector for the deck | Print worksheets. That's it. |

## OBJECTIVES

- Explain how an AI learns: **input** → **patterns** → **output** — prediction from data, not magic. *(Exit Ticket #1)*
- Name who a dataset serves well and who it leaves out; link that gap to real AI bias. *(Gap Report; Exit #2)*
- Discuss data gaps collaboratively: repeat a partner's idea, then add to it. *(Repeat, Then Add; Exit Ticket #3)*

## STANDARDS ALIGNMENT

CSTA 2-IC-21 · ISTE 1.1.d · CCSS ELA SL.6.1 · Full text: Teacher Guide, p.3

## DIFFERENTIATION

- **Support:** Grey sentence starters on the worksheet; landmark-card tip in the Teacher Guide, p.2.
- **Extension:** Design a 12-card playlist that serves every request on the sheet, then defend the trade-offs.
- **ELL:** Fit marks are checked boxes, not sentences; invented cards may be in any language (that fixes a real gap).
- **No-device classrooms:** Built in — the lesson is paper, pairs, and one loud imaginary music app.

## WARM-UP — THE ONE-PLAYLIST APP (7 MIN)

You are **AutoMix**, the school's new music app. Your playlist holds one kind of song: loud, fast party tracks. Take 3–4 requests and serve every one with total confidence:

*"Something chill to study to?"* → "Here's 'TURBO TURBO.' It is not chill. It's what I've got."

*"A song for grandma's birthday dinner?"* → "Got it. 'VOLUME WAR.' Grandma loves bass drops, probably."

*"A slow song for the last dance?"* → "'NEON STAMPEDE,' played at... the exact same speed."

Then ask: **"Is AutoMix bad at picking songs — or is something else broken?"** Fish for it: the app is doing its job perfectly. The *playlist* is the problem.

*If nobody gets there: "Could AutoMix play a slow song if it had one?" Yes — the app works. So what's missing?*

**Bridge:** "(If your class did the genie lesson: it granted what you SAID.) Today: where an AI's answers even come from. Everything an AI 'knows' comes from a playlist of examples — its training data. It can remix the playlist, but it can't serve what was never in it. Today you'll BE the app."

**Transition:** "So what IS in an AI's playlist, and what does it do with it? Three words on the board."

## DIRECT INSTRUCTION — NO MAGIC IN THE MACHINE (8 MIN)

**INPUT — the playlist.** Training data: every example the AI studied before you met it. Nobody types "knowledge" in.

**PATTERNS — the training.** It gets good at ONE thing: guessing what usually goes with what. No understanding — patterns.

**OUTPUT — the mix.** It predicts what best matches those patterns. A prediction, not a lookup — confident and still off.

**Say it plainly** (board: INPUT → PATTERNS → OUTPUT, one word per box)**:** "No magic in the machine. An AI doesn't KNOW things — it predicts things, from its playlist. So two rules run everything: a GAP in the playlist is a gap in the AI, and a LEAN in the playlist leans every answer. That lean has a name: bias."

**Quick check:** "A photo AI's playlist is almost all golden retrievers. A husky walks in — what's it going to call it?" (Golden retriever — its best guess. Gap in, gap out.)

*If a student says "husky": "Has this AI ever seen one? It can only pick from the patterns it has."*

**Transition:** "Hand out Playlist Check. Pair up. Read the grey box at the top together — it names your two jobs."

## ACTIVITY — PLAYLIST CHECK (20 MIN)

1. **Setup (3 min):** Partner A is the **App**, Partner B the **Request Line**; the 12 cards ARE the app's training data. Rules: serve only from the playlist, never say no, check ONE Fit box honestly — each on their OWN sheet.
   *If there's an odd number: a trio — two Request Lines, one App, swap twice.*
   *Transition:* "Request Line, read #1 out loud. App — you can't say no."
2. **Serve the Request Line (9 min):** Six requests (Priya, Nia, Tomás, Keisha, Malik, Jordan): App names the card, both agree on one Fit box and check it. Swap at #4; end with one live request from a neighboring pair.
   *If a pair can't agree on the Fit: each checks their own — disagreement is data. Save it for the debrief.*
   *Transitions:* (#3) "Swap jobs: new App, new Request Line. On to #4." (#7) "Turn to the pair beside you. Make up ONE request for their app — they can't say no." (after #7) "Pens on page 2. Say who it let down."
3. **Playlist Gap Report (5 min):** Each student writes who the playlist serves well and who it leaves out. Rule, enforced out loud: *repeat your partner's idea, then add to it* — both go on the Repeat, Then Add lines.
   *If a pair writes "it needs more songs": ask WHO asked for something that wasn't there.*
   *Transition:* "Three minutes. Fix the playlist — invent the cards that are missing."
4. **Fix the Playlist (3 min):** Invent new cards that fix the biggest gaps — at least 2. Push past "add one slow song": which requests still have nothing?
   *If every new card is a slow song: "Tomás still has nothing. What does he need?"*
   *Transition:* "Pens down. Eyes up. Who did your playlist let down? Hands."

## DEBRIEF — THE REAL PLAYLISTS (5 MIN)

**2–3 pairs** name one gap (1 min). Three real playlists, Teacher Guide p.2 (3 min). Big question (1 min): **"Nobody put bad songs in your playlist — so where did the bias come from?"**

*Answer: what got collected, and what didn't. "On purpose?" Usually not — teams grab the easiest data to get.*

**Transition:** "Back to your own sheet. The bottom of page 2 is yours alone — five minutes."

## EXIT TICKET (5 MIN)

Worksheet p.2, solo: **#1** finish **input** → **patterns** → **output** + what an AI does with its playlist; **#2** diagnose a lopsided homework-helper AI using "playlist"; **#3** one gap a partner spotted first. Key: Teacher Guide p.2.

*If a student writes "the AI learns": "Learns from what? Say where the patterns came from."*